CLASS 10 MATHEMATICS • CHAPTER 1

Real Numbers

Exhaustive point-by-point presentation covering every concept, theorem, and formula directly from the textbook.

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Continuing the Exploration of Real Numbers

What We Studied Earlier

- Explored the world of real numbers and encountered irrational numbers in Class IX.
- Learned about the union of rational and irrational numbers making up real numbers.

Path Ahead in Chapter 1

- **Euclid's Division Algorithm:** Properties of positive integers and computing HCF.
- **Fundamental Theorem of Arithmetic:** Prime factorisation, uniqueness, and proofs of irrationality.

Section 1.1 Live Check

Q1. Can you recall what makes a number rational versus irrational? Give two quick examples of each.

Q2. What are the two core pillars or main theorems we are going to explore for positive integers in this chapter?

Q3. Why is prime factorisation considered a powerful tool in arithmetic?

Building Numbers from Primes

Multiplication Perspective

If we take any collection of prime numbers (e.g., 2, 3, 7, 11, 23) and multiply them repeatedly, we generate infinitely many positive integers.

$$7 \times 11 \times 23 = 1771$$
$$2^2 \times 3 \times 7 \times 11 \times 23 = 21252$$

Factorisation Perspective

- Doing the opposite: breaking down any large composite number into prime factors using a **factor tree**.
- Example from text: 32760 factorised gives $2^3 \times 3^2 \times 5 \times 7 \times 13$.
- Leads us directly to the fundamental foundational theorem of integer arithmetic!

Section 1.2 (Part 1) Live Check

Q1. Can every natural number be obtained by multiplying prime numbers? What do you think?

Q2. How do we construct a factor tree for a number like 32760?

Q3. Is there any composite number that cannot be expressed as a product of prime powers?

Theorem 1.1 & Historical Context

Theorem 1.1 (Fundamental Theorem of Arithmetic)

Every composite number can be expressed (factorised) as a product of primes, and this factorisation is **unique**, apart from the order in which the prime factors occur.

Uniqueness Meaning

$2 \times 3 \times 5 \times 7$ is regarded as the exact same factorisation as $3 \times 5 \times 7 \times 2$. Written in ascending order: $x = p_1 p_2 \ldots p_n$ where $p_1 \leq p_2 \leq \ldots \leq p_n$.

Carl Friedrich Gauss (1777–1855)

Often called the **'Prince of Mathematicians'**. While Euclid recorded early versions in Elements, Gauss gave the first correct proof in *Disquisitiones Arithmeticae*.

Theorem 1.1 Live Check

Q1. State the Fundamental Theorem of Arithmetic in your own words.

Q2. Why do we say factorisation is unique 'except for the order of factors'? Give an example.

Q3. Who gave the first correct proof of this fundamental theorem?

Applications: Checking End Digits & HCF/LCM

Example 1: Does 4^n end with 0?

If 4^n ends with 0, it must be divisible by 5 (prime factor 5). But $4^n = (2)^{power}$, whose only prime factor is 2. By uniqueness, 5 cannot appear. Thus, **no such n exists**.

HCF & LCM Rules

- **HCF(a, b):** Product of smallest power of each common prime factor.
- **LCM(a, b):** Product of greatest power of each prime factor involved.
- **Property:** HCF(a, b) × LCM(a, b) = a × b.

Applications Live Check

Q1. Why must a number ending in zero have 5 as a prime factor?

Q2. How do we find the HCF of two numbers using their prime factorisations?

Q3. Is the product of HCF and LCM always equal to the product of three numbers? (Think!)

Prime Factorisation in Action

Example 3 (96 & 404)

- $96 = 2^5 \times 3$, $404 = 2^2 \times 101$
- HCF = $2^2 = 4$
- LCM = $(96 \times 404) / 4 =$ **9696**

Example 4 (6, 72, 120)

- $6 = 2 \times 3$, $72 = 2^3 \times 3^2$, $120 = 2^3 \times 3 \times 5$
- HCF = $2^1 \times 3^1 = 6$
- LCM = $2^3 \times 3^2 \times 5^1 = 360$
- *Remark:* HCF × LCM ≠ product of 3 numbers!

Examples Live Check

Q1. How can we easily find the LCM of two numbers once their HCF is known?

Q2. For three numbers p, q, and r, is HCF × LCM equal to their product? What did Example 4 show?

Q3. What are the smallest powers of common prime factors used for determining HCF?

Irrational Numbers & Divisibility Lemma

Theorem 1.2 (Prime Divisibility Property)

Let p be a prime number. If p divides a^2, then **p divides a**, where a is a positive integer.

What is an Irrational Number?

A number 's' is irrational if it cannot be written in the form p/q, where p and q are integers and q ≠ 0. Examples: √2, √3, √5, π, -√2/√3, 0.101101110...

Section 1.3 (Part 1) Live Check

Q1. What is the definition of an irrational number?

Q2. State Theorem 1.2 in your own words regarding prime divisibility.

Q3. Can you name three standard irrational numbers studied in Class IX?

Proof by Contradiction: √2 and √3

Method: Proof by Contradiction

- Assume the opposite: suppose √2 is rational (= a/b where a, b are coprime).
- Square both sides: $2b^2 = a^2 \Rightarrow$ 2 divides a^2.
- By Theorem 1.2, 2 divides a (let a = 2c).
- Substitute back: $b^2 = 2c^2 \Rightarrow$ 2 divides b.
- Contradiction! (a and b share factor 2, but are coprime). Thus, √2 is irrational.

Generalising for √3 (Example 5)

- Similarly assume √3 = a/b (coprime).
- $3b^2 = a^2 \Rightarrow$ 3 divides a. Let a = 3c.
- $b^2 = 3c^2 \Rightarrow$ 3 divides b.
- Contradiction proves **√3 is irrational**. (Applies to all √p where p is prime).

Theorem 1.3 Live Check

Q1. What logical technique is used to prove that √2 is irrational?

Q2. Why do we assume a and b are coprime integers at the beginning of the proof?

Q3. What contradiction arises that completes the proof for √3?

Combinations of Rational & Irrational Numbers

Example 6: 5 - √3 is Irrational

- Assume rational: 5 - √3 = a/b.
- Rearrange: √3 = 5 - (a/b) = (5b - a)/b.
- Right side is rational, so √3 must be rational.
- Contradicts the fact that √3 is irrational. Hence proved!

Example 7: 3√2 is Irrational

- Assume rational: 3√2 = a/b.
- Rearrange: √2 = a / (3b).
- Since 3, a, b are integers, a/(3b) is rational, implying √2 is rational.
- Contradiction! Thus **3√2 is irrational**.

Examples 6 & 7 Live Check

Q1. How do we isolate the pure irrational term when testing expressions like 5 - √3?

Q2. What general rule governs the sum/difference and product/quotient of a rational and irrational number?

Q3. Why does assuming 3√2 is rational lead to an impossible statement?

Worked Exercise Highlights

Problem 1: Express 3825 as Primes

- 3825 ends in 5, divisible by 5: 5 × 765
- 765 = 5 × 153
- 153 = 3 × 51 = 3 × 3 × 17
- **Answer: $3^2 \times 5^2 \times 17$**

Problem 2: Prove $3 + 2\sqrt{5}$ is Irrational

- Assume rational: $3 + 2\sqrt{5} = a/b$.
- $2\sqrt{5} = (a/b) - 3 = (a - 3b)/b$.
- $\sqrt{5} = (a - 3b) / (2b)$.
- RHS is rational, but $\sqrt{5}$ is irrational. Contradiction! **Proved.**

Exercise Highlights Live Check

Q1. How do we test divisibility by small primes when prime factorising numbers like 3825?

Q2. In proving 3 + 2√5 is irrational, why do we isolate the radical term (√5)?

Q3. Can you explain why circular track problems (like Sonia and Ravi running) use LCM?

Chapter Recap & Checklist

Core Theorems

- **Fundamental Theorem of Arithmetic:** Unique prime factorisation of composite numbers.
- **Theorem 1.2:** If prime p divides a^2, then p divides a.

Methods & Notes

- **Irrationality Proofs:** Proven for √2, √3, √5 using proof by contradiction.
- **Note on 3 Numbers:** HCF(p,q,r) × LCM(p,q,r) ≠ p × q × r. Special formulas apply!

THANK YOU

"Mathematics is the music of reason." — James Joseph Sylvester

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